

Kashmith Samaranayaka

Senior Software Engineer · AI/ML

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Professional Summary

Software Engineer specializing in **AI/ML and backend systems**, with production experience building LLM-powered and agentic applications, MLOps pipelines, and scalable infrastructure. **Open-source core contributor to the Jaseci stack**, a major developer of the **jac-scale** Kubernetes deployment plugin and contributor to the **jaclang** language. Co-author of **two IEEE-published papers** on deep learning. Graduate of the **University of Moratuwa**.

Technical Skills

Languages: Python, Go, Jac (jaclang), SQL, Bash

AI/ML & GenAI: LLMs, RAG, Agentic AI (LangGraph), Fine-tuning (Unsloth), NLP, Deep Learning

ML Frameworks: PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face

MLOps & Infra: Docker, Kubernetes, MLflow, Apache Airflow, CI/CD

Backend & Data: FastAPI, Jaseci, REST APIs, PySpark, Kafka, Redis Streams, Vector Databases (Chroma)

Experience

AI/ML Engineer

April 2024 - Present

Jaseci Lab, Ann Arbor, MI, USA

Remote

- Core contributor to the open-source **Jaseci stack**, a major developer of the **jac-scale** Kubernetes deployment plugin and contributor to the **jaclang** language.
- Developed and maintained backend infrastructure for **Tobu**, an AI-powered personal memory platform with integrated LLMs serving **1,000+ users**; took the system to production fast enough to support the company's **\$1.5M raise**.
- Built scalable PySpark data pipelines for user analysis and feature optimization, and shipped ML features for image analysis and conversation processing.

Data Scientist

April 2024 - June 2024

Utech Technologies (Pvt) Limited, Wattala, Sri Lanka

Remote

- Designed and deployed an end-to-end energy forecasting pipeline with MLflow experiment tracking and model versioning for reproducible, production-ready forecasts.
- Automated ML workflows with Apache Airflow for data ingestion, model retraining, and deployment.
- Built production API services with FastAPI and Docker for real-time energy prediction and dashboard visualization.

Application Engineer Intern

January 2023 - July 2023

Synopsys Lanka (Pvt) Ltd, Colombo, Sri Lanka

On-site

- Developed an NLP-based search engine with LSTM models for automated email classification.
- Automated large-file compilation and notification systems using Python, Bash, and CSH scripting.
- Managed 17 processor designs through custom automation scripts for daily performance assessment and resource optimization.

Education

B.Sc. Engineering (Hons.) in Electrical Engineering

Jan. 2020 - July 2024

University of Moratuwa

Moratuwa, Sri Lanka

- Overall GPA: 3.31 (Second Upper Class)** · Washington Accord-accredited program
- Dean's List – 2nd semester

Open-Source Contributions

Jaseci Stack: Core Contributor (jac-scale, jaclang)

- Core developer of **jac-scale**, the deployment plugin that scales identical local Jac code to Kubernetes production, auto-provisioning databases, Redis caching, and authentication with zero configuration.
- Contributor to **jaclang**, an AI-native language featuring graph-based Object-Spatial Programming with native Python, JavaScript, and C interoperability.

Projects

Industry & AI Projects

JacHammer - AI Agentic Software-Generation Platform

- Backend developer on an agentic AI platform that generates and deploys full-stack jaclang applications end-to-end, orchestrating LLM-driven code generation, build, and one-click deployment.

Tobu Lite - AI Image-to-Video Generation

- Built the image-to-video feature for the Tobu platform, a generative media pipeline integrating Google's **Veo 3** model that lets users create videos from their photos using selectable templates.

Deep Research Assistant - Agentic AI Application

- Built a production-grade agentic app for analytical question answering using LangGraph for multi-step reasoning, with web search and Chroma vector retrieval, deployed via Docker with a Streamlit interface.

Einstein Pro - AI Patent Generation & Document Management Platform

- Backend developer for an AI platform that generates patents and manages documents, including a RAG-based chatbot over a vector database for fast, accurate retrieval.

FriendZone - AI-Powered Memory Sharing Platform

- Full-stack AI platform to capture, organize, and share memories via conversational interfaces, with automatic image analysis (who/what/where/when) using the OpenAI API; backend built on jaclang and jac-cloud.

Self-Adaptive NILM using Deep Learning (Research)

- Designed WaveNet/CNN models with ensemble methods for appliance-level power prediction; applied pseudo-labelling and transfer learning for self-adaptation to appliance ageing without retraining.
- Co-authored and published the work at the **2024 IEEE World AI IoT Congress** (see Publications).

Publications

M. S. K. Nisakya, S. M. L. Arampola, W. A. Yasodya, S. Kumarawadu, V. Logeeshan, C. Wanigasekara. "Self-Adaptive Non-Intrusive Load Monitoring Using Deep Learning." *2024 IEEE World AI IoT Congress*, 2024. 

W. A. Yasodya, S. M. L. Arampola, M. S. K. Nisakya, V. Logeeshan, S. Kumarawadu, C. Wanigasekara. "Self-Adaptive Deep Learning Framework for Non-Intrusive Load Monitoring: Addressing Aging Appliance Challenges With Transfer Learning and Pseudo Labeling." *IEEE Access*, vol. 13, pp. 106524–106539, 2025. 

Professional Certifications

Machine Learning Specialization 
DeepLearning.AI & Stanford University

TensorFlow: Advanced Techniques Specialization 
DeepLearning.AI

Certified Kubernetes Administrator (CKA) (*In Progress*)
Udemy

Machine Learning Engineering for Production (MLOps) Specialization 
DeepLearning.AI

Data Structures and Algorithms Specialization
University of California San Diego

AWS Solutions Architect – Professional (*In Progress*)
Udemy